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c0055 Chapter 11

Executive functions

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s0010 INTRODUCTION

p0010 Executive functions (EFs) serve as an umbrella term to encompass the set of higher-order cognitive abilities that are necessary to peruse and achieve a goal. These functions are what enable us to understand complex or abstract concepts, solve problems we never encountered before, plan our next vacation, and manage our relationships. However, despite its importance, the executive system has been traditionally quite difficult to define. While it is intuitive to think how a patient with a short-term memory problem might manifest in the clinic, it is harder to predict the specific manifestation of an executive-impaired patient since there is no single behavior that can be directly tied to EF.

p0015 For decades, EFs have been associated with the frontal lobes. This association originated in the famous case of Phineas Gage, who was working on a railroad construction site when in 1848 a large iron rod passed through his left frontal lobe. Phineas Gage survived this accident but his behavior and personality were dramatically altered, providing the first documented evidence for the complexity of EFs and their neural basis (Ratiu et al., 2004).

p0020 Today, EFs are best described as a complex set of cognitive abilities that includes working memory, inhibitory control, cognitive flexibility, planning, reasoning, and problem solving (Cicerone et al., 2000; Kennedy et al., 2008; see Table 11.1 for definitions).

p0025 The executive system manages and controls other cognitive abilities (e.g., attention and memory) and allows individuals to alter their overlearned behavioral patterns when they become unsatisfactory (Van der Linden et al., 2000). It also enables adaptation to novel and complex everyday life situations (Collette et al., 2006). Consequently, deficits in EF have dramatic impacts on everyday life, which include poor goal-directed behavior and impaired social functioning (Hanks et al., 1999; Busch et al., 2005) among others (see Table 11.2).

This chapter begins with a description of the models proposed to explain EF and the association between EF and frontal lobes. Next, it reviews lesion and neuroimaging evidence that has improved our understanding of the neuroanatomy of EF, in particular inhibitory control, cognitive flexibility, planning, reasoning, and problem solving. Later, the main neuropsychologic tools used to assess EF are outlined, with particular attention paid to the contribution that ecological assessment can bring to the field. Then, the main approaches in rehabilitation and training are reviewed, with a discussion of how EFs evolve and change through the life span. Finally, the interplay between genetics and EF is explored. p0030

EF MODELS s0015

Throughout the years, different frameworks were proposed to conceptualize the executive system and the way it operates. This section describes some of the most prominent models. p0035

Luria (1973) was the first to conceptualize an EF model. He considered problem-solving behavior depending on a number of overriding skills, or EFs, which rely on the correct functioning of the frontal lobes. He described the major components of the EF system as anticipation (setting realistic expectations, understanding consequences), planning (organization), execution (flexibility, maintaining set), and self-monitoring (emotional control, error recognition). p0040

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t0010 ***Table 11.1***

Main EF process with their definition (working memory, inhibitory control, cognitive flexibility, planning, reasoning, problem solving).

EF process	Definition
Working memory	Holds information online during the execution of other cognitive functions, such as taking notes during a lecture or paraphrasing information we hear or read about. Working memory has been investigated by tasks such as the *n*-back task, where a series of stimuli is presented and subjects have to indicate if the current stimulus matches one that appeared earlier in the series. An increase in *n* increases working memory load, making the task more demanding. Since a decision needs to be made after each stimulus, the *n*-back task requires a continuous online updating of information in working memory
Inhibitory control	Interacts with working memory and cognitive control to monitor adaptive behaviors such as withholding or suppressing a response that is no longer relevant (e.g., go/no-go task) or suppressing the recovery of irrelevant information from memory (e.g., directed forgetting)
Cognitive flexibility	Cognitive flexibility is a hallmark of human thought, enabling the ability to adapt in the face of environmental change and to generate new ideas that drive innovation and promote growth and discovery (Badre and Wagner, 2006). Allows switching from one task to another and is linked with inhibitory control and working memory. It has been widely studied with the Wisconsin Card Sorting Test (Milner, 1963) and requires skills such as conceptualizing the sorting criteria, making hypotheses about the criteria, performance monitoring, and using feedback to modify the strategy once the rule has changed
Planning	Planning is a higher-level cognitive function that includes EF processes involved in the formulation, evaluation, and selection of actions required to attain a goal. Planning ability has been studied using various tasks, including the Tower of London (Shallice, 1982). This task requires the subject to rearrange beads on three vertical rods in order to match a model in as few moves as possible. The Tower of London relies on working memory to reach intermediate goals, while the final goal is maintained in memory
Reasoning	Reasoning is the core of the generalization and abstraction processes that enable concept formation and creativity
Problem solving	The process of working through details of a problem to reach a solution. Problem solving may include mathematical or systematic operations and can be a gage of an individual's critical thinking

p0045 Later, Lezak (1995) defined EFs as the mental capabilities necessary for formulating an objective, planning, and implementing actions to achieve that objective, and then intervening in the performance of complex tasks. These functions act when behavioral control processes are required and depend on three cognitive actions: shifting or changing between different tasks or mental sets, inhibiting irrelevant automatic responses, and updating mental representations held in working memory (Miyake et al., 2000; Van der Linden et al., 2000). The concept of self-regulation of behavior is the ability to regulate behavior according to internal goals and constraints (Goldberg and Podell, 2000). This arises from the ability to hold a mental representation and to use this information to inhibit inappropriate responses (Levine et al., 1998), implying the ability to modify behavior while taking the consequences of actions into account.

p0050 Stuss and Benson (1986) proposed another model of EF, which is closely related to the model proposed by Lezak that was inspired by Luria's studies of EF. According to their model, EF processes include initiation of behavior, followed by planning, sequencing, and organization. These three subprocesses are similar to Luria's components of planning and execution. However, Stuss later placed EF in the middle of a hierarchical framework (Stuss, 1991). According to this model, EFs receive input from lower-level processes (e.g., attention, memory, language, perception), as well as higher-level metacognitive processes.

Sohlberg and Mateer (1987) and, more recently, p0055 Mateer (1999) and Sohlberg and Mateer (2001) conceptualized a clinical model of EF involving six components: (1) initiation and drive (activation or starting of a cognitive system), (2) response inhibition (stopping automatic or prepotent response tendencies), (3) task persistence (maintaining a behavior until task completion), (4) organization (organizing and sequencing of information), (5) generative thinking (creating multiple solutions to a problem and thinking in a flexible manner), and (6) awareness (monitoring and modifying one's own behavior). They use this model to guide their observations, assessment, and management plan.

More recently, Grafman (2002) introduced the struc- p0060 tured event complex (SEC) framework, suggesting that the prefrontal cortex (PFC) stores unique forms of hierarchical knowledge (Grafman, 2002) that, when activated, appear as EFs. An SEC is a goal-oriented set of events structured in sequence and represents thematic

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t0015 *Table 11.2*

Impact of EF in everyday life.

Life domain	Impact of EF
Mental health	EF are impaired in several mental disorders such as: Attention deficit hyperactivity disorder (Lui and Tannock, 2007) Schizophrenia (Barch, 2005) Depression (Taylor Tavares et al., 2007) Conduct disorder (Fairchild et al., 2009) Addiction (Baler and Volkow, 2006)
Physical health	Poor EFs are associated with: Obesity (Crescioni et al., 2011) Substance abuse (Miller et al., 2013) Poor treatment adherence (Riggs et al., 2010)
Quality of life	Individuals with good EFs have a better quality of life (Brown and Landgraf, 2010; Davis et al., 2010)
School success	EFs are good predictors of math and reading abilities (Blair and Razza, 2007)
Job success	Poor EFs are associated with lower work productivity and more difficulty in finding and keeping a job (Bailey, 2007)
Marital life	Poor EFs are associated with an impulsive partner (Eakin et al., 2004)
Public security	Poor EFs are associated with crime and violence (Broidy et al., 2003; Denson et al., 2011)

knowledge, morals, abstractions, concepts, social rules, event features, event boundaries, and grammars. The stored characteristics of these representations form the bases for the strength of representation in memory and the relationships between SEC representations.

p0065 Lastly, two EF models were proposed that relied heavily upon functional neuroimaging evidence. First, Badre and D'Esposito (2007) considered the PFC to be crucial to flexible and organized action. These researchers provided evidence of the supposed PFC rostrocaudal hierarchical organization of control. A systematic posterior-to-anterior gradient was evident within the PFC depending on the manipulated level of representation. Furthermore, across four functional MRI (fMRI) experiments, activation in PFC subregions was consistent with the sub- and superordinate relationships that define an abstract representational hierarchy. In addition to providing further support for a representational hierarchy account of the rostrocaudal gradient in the PFC, these data provide important empirical constraints on current theorizing about control hierarchies and the PFC. Second, Koechlin and Summerfield (2007) proposed a theory that draws upon concepts from information theory to describe the architecture of executive control in the lateral PFC. Supported by evidence from brain imaging in human subjects, the model proposes that action selection is guided by hierarchically ordered control signals, processed in a network of brain regions organized along the anterior–posterior axis of the lateral PFC. The theory clarifies how executive control can operate as a unitary function, despite the requirement that information be integrated across multiple, functionally specialized prefrontal regions.

Although the models described here provide somewhat differing views about the organization of the EF system, they all agree that EFs involve several different subfunctions, mostly subserved by the PFC. Lesion and neuroimaging studies are particularly valuable for identifying the neural bases of EF and to adjudicate which model best predicts the EF performance. p0070

NEUROANATOMY OF EF s0020

As noted earlier, the PFC is the brain region most commonly associated with EF. The PFC comprises more than 30% of the entire complement of cortical cells and is the most recently evolved brain region (Semendeferi et al., 2002). It is also the part of the cortex that is more highly developed in humans than in other primates (Hoffmann, 2013). There are three broad PFC subdivisions: the medial part, the dorsolateral part, and the orbitofrontal region (Ongür et al., 2003; see Fig. 11.1). The PFC receives input projections from other neocortical areas, such as parietal and temporal regions. The PFC also receives information from the hippocampus, the cingulate cortex, the substantia nigra, and the thalamus, primarily from the medial dorsal nuclei. The PFC sends back projections to the medial dorsal nuclei as well as to the amygdala, the septal nuclei, the basal ganglia, and the hypothalamus (McGarry and Carter, 2017), and therefore is highly interconnected with other cortical and subcortical structures. p0075

EF lesion mapping studies s0025

Much of the evidence for the anatomic association between PFC and EF comes from studying individuals who suffered a brain injury. Neuropsychologic studies of patients with focal brain lesions provide an opportunity to study specific lesion-deficit associations. Importantly, lesion studies, unlike neuroimaging studies, allow the examination of causal—rather than correlational—hypotheses. p0080

PFC AND UNDERLYING WHITE MATTER TRACTS s0030

Due to its position in the skull, the PFC is frequently damaged after traumatic brain injury (TBI) (Levin p0085

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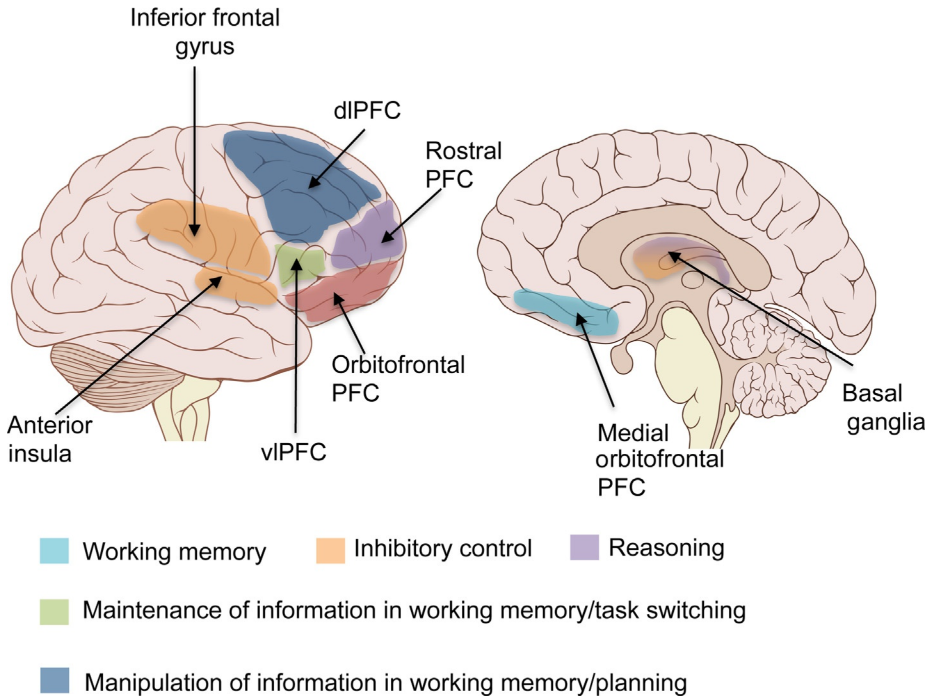

f0010 **Fig. 11.1.** Brain areas associated with different EF based on lesion and/or neuroimaging studies. Brain image is adapted from work by Patrick J. Lynch, medical illustrator; C. Carl Jaffe, MD, cardiologist Creative Commons Attribution 2.5 License 2006.

et al., 1987). Studies involving individuals with penetrating TBI have shown that the dorsolateral PFC (dlPFC) and orbitofrontal cortex (OFC) are associated with EF impairments (Barbey et al., 2012). Individuals with larger lesions in the PFC may present with *dysexecutive syndrome* (Baddeley and Wilson, 1988), which impairs cognition, emotion, social behavior, and goal-oriented behavior (Gansler et al., 1996; Brooks et al., 1999; Cicerone et al., 2000). Damage to the OFC and anterior cingulate cortex (ACC) can lead to impairments of social behavior and motivated behavior. Individuals with lesions to the dlPFC show higher-order cognitive problems involving goal-directed behaviors (Al-Adawi et al., 1998), cognitive control (Larson et al., 2006), inhibition (Picton et al., 2007), planning (Cicerone and Wood, 1987), and working memory (Barbey et al., 2013b). As these deficits impair patients' everyday activities and autonomy, they are important predictors of negative outcomes (Vilkki et al., 1994; Reid-Arndt et al., 2007).

p0090 However, nonfrontal brain areas are also found to be associated with EF. For example, it has been recognized that EFs rely on several other brain areas that are closely linked with the PFC and form larger executive neural networks (Andrés, 2003; Collette et al., 2006; Champod and Petrides, 2007). For instance, the PFC is strongly linked with the limbic region of the medial temporal lobe (such as hippocampus, amygdala), and these connections are crucial for mnemonic interactions and emotional response regulation (Barbas and Zikopoulos, 2007; Petrides et al., 2007). The PFC and the hippocampus are both connected to the nucleus accumbens, which plays a crucial role in integrating cortical and limbic information into goal-directed behavior (Pennartz et al., 1994).

More recently, Cristofori et al. (2015) showed that white matter (WM) plays a crucial role in EF functionality. In particular, damage to the anterior corona radiata and superior longitudinal fasciculus was associated with a poorer functional recovery in a verbal fluency task, which relies upon the fronto-temporo-parietal network. On the other hand, the Trail Making and 20 Questions tasks, which are linked to more focal left frontal damage, were better predicted by PFC lesions. WM damage can place additional burden on recovery by impairing signal transmission between cortical areas within a large functional brain network such as the one involved in EF. p0095

Taken together, these studies suggest that EF relies mostly on PFC neural networks; however, other cortical (e.g., parietal and temporal cortex) and subcortical (e.g., WM such as anterior corona radiata and superior longitudinal fasciculus) brain regions connected to the PFC can influence the correct functioning of EF. p0100

Working memory s0035

Neural activity in dlPFC stores and maintains working memory representations (Goldman-Rakic, 1995). The p0105

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evidence for this theory relies on two main findings from nonhuman primates. First, neuronal activity in monkey dlPFC neurons selective for the spatial location of a remembered target persists during location retention (Funahashi et al., 1989). Second, experimental lesions of the dlPFC impair the monkeys' ability to remember locations in the contralesional visual field (Funahashi et al., 1993). However, neuroimaging studies usually fail to find neural activity in the human dlPFC during similar working memory tasks (Courtney et al., 1998; Srimal and Curtis, 2008).

p0110 Neuropsychologic studies of working memory attempted to resolve this controversy. For example, D'Esposito and Postle (1999) performed an analysis of reports investigating working memory performance in patients with PFC lesions. The authors focused on studies involving verbal and spatial span and delayed response tasks. The studies reviewed indicate that an intact lateral PFC is necessary for normal performance in working memory, especially when the task involves distraction or task difficulty is increased in other ways. None of the studies reviewed indicated that a span deficit was associated with PFC lesion (D'Esposito and Postle, 1999).

p0115 More recently, accumulating evidence indicates that the OFC mediates executive control functions underlying the coordination of multiple working memory processes (for meta-analytic reviews, see Wager and Smith, 2003b; Owen et al., 2005). According to this framework, the OFC is engaged when problems involve more than one cognitive process, such as when one cognitive process is not sufficient and the integration of two or more separate cognitive processes is required to achieve the behavioral goal. The *n*-back task is an ideal example of such a behavioral challenge. In this task participants are asked to monitor a target stimulus and to indicate when the current stimulus is the same as the one presented *n* trials before. This task requires the participant to simultaneously monitor a series of stimuli, maintain activation of recently processed and potentially relevant items, discard recently processed but irrelevant information, and make comparisons between various items in the series to identify a correct match. Multiple cognitive processes can be carried out successfully only if they are coordinated, and it is suggested that the coordination of information processing and information transfer between multiple processes is regulated by the OFC.

p0120 Barbey et al. (2011) characterized working memory function in a group of patients with OFC lesions. Using a large sample of patients with OFC damage ($n=24$) and a wide-ranging assessment of cognitive function, they reported that: (1) medial OFC lesions were associated with reliable deficits in executive control functions underlying the joint maintenance, manipulation, and monitoring of information in working memory, particularly at high levels of cognitive load; (2) medial OFC lesions were not associated with pervasive deficits on tests of working memory maintenance or manipulation; (3) medial OFC lesions were not associated with reliable deficits in tests of verbal or spatial reasoning; and (4) lateral OFC lesions did not yield impairment in working memory function. Together, these results indicated that the neural architecture of the medial OFC is necessary for the coordination of working memory maintenance, manipulation, and monitoring processes.

Another recent study (Sepulcre et al., 2009) investigated the WM damage associated with working memory p0125
impairments in multiple sclerosis patients. Using voxel-based lesion–function mapping and fMRI analyses, the authors identified critical WM tracts involved in working memory measured with a Paced Auditory Serial Addition Test (Gronwall, 1977). The WM tracts involved in this kind of task included the cingulum, the parietofrontal pathways, the thalamocortical projections, and the right cerebellar WM. This study confirmed that pathways connecting the parietal lobes with the lateral PFC and ACC, as well as the thalamic radiations, play a crucial role in working memory processes, as all these regions are known to mediate working memory processes (Smith and Jonides, 1998; Baddeley, 2003; Audoin et al., 2005).

Neuropsychologic patient data further complement p0130
the nonhuman primate (for example, Goldman-Rakic, 1995; Fuster, 1997; Levy and Goldman-Rakic, 2000) and functional neuroimaging literature (for example, Baldo and Dronkers, 2006; Volle et al., 2008; Tsuchida and Fellows, 2009), which have traditionally focused on dlPFC and ventrolateral prefrontal cortex (vlPFC) contributions to working memory. Neuropsychologic data, however, have failed to corroborate neuroscience evidence implicating the ventrolateral PFC in the maintenance of information in working memory, suggesting that this region may not be a necessary component of the neural systems mediating human working memory.

INHIBITORY CONTROL s0040

It is recognized that the PFC serves as a source of inhib- p0135
itory control over other brain areas. A prevalent view is that certain PFC regions are specialized for inhibitory control, such as the right inferior frontal gyrus (Tabibnia et al., 2011). More recent views considered a more unified framework for understanding inhibitory control in the broader context of PFC function.

Most of the evidence regarding functional-anatomic p0140
networks of cognitive inhibition has come from functional imaging studies (see section "Working memory").

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Few lesion studies have supported a causal role for different sectors of the PFC in cognitive control (Stuss et al., 2000, 2001).

p0145 These first studies used isolated neuropsychologic tasks, a limited number of patients, and focused on particular a priori hypothesized sectors of the PFC, limiting their neuroanatomic conclusions. Recently, Gläscher et al. (2009) used voxel-based lesion–symptom mapping in a large sample of individuals with focal lesions (including PFC) together with a comprehensive battery of neuropsychologic tasks to measure cognitive inhibition/control. The findings revealed that cognitive control (response inhibition, conflict monitoring, and switching) was associated with lesions to the dlPFC and ACC.

s0045 Cognitive flexibility

p0150 Lesions to different prefrontal areas result in different deficits of cognitive flexibility. Studies have shown, for example, that damage to the OFC impairs reversal learning, but not attentional set-shifting (McAlonan and Brown, 2003; Boulougouris et al., 2007), while damage to the lateral PFC impairs shifting of attentional sets but not reversal learning (Owen et al., 1991; Bissonette et al., 2008). Recently, the proposed unique role of the OFC in reversal learning (based on the idea that the OFC is a rapidly flexible associative learning area) is under discussion and alternative views have been presented. In fact, recent findings suggest that the OFC is crucial for signaling outcome expectancies, and therefore changing behavior when confronted with unexpected outcomes (Schoenbaum et al., 2009). Other studies indicate that impaired reversal learning in rhesus monkeys is only observed following aspiration but not stereotaxic/focal OFC lesions (Rudebeck et al., 2013), suggesting that reversal learning does not depend on an intact OFC but instead on intact communication between other prefrontal areas and more caudal structures.

p0155 Neuroscience has made remarkable progress in understanding the architecture of human intelligence, identifying a distributed network of brain structures that support goal-directed, intelligent behavior. However, the neural foundations of cognitive flexibility and the adaptive aspects of intellectual function remain to be well characterized. Barbey et al. (2013a) reported a human lesion study ($n = 149$) that investigated the neural bases of key competencies of cognitive flexibility and examined their contributions to a broad spectrum of cognitive and social processes, including psychometric intelligence (Wechsler Adult Intelligence Scale), emotional intelligence (Mayer, Salovey, Caruso Emotional Intelligence Test), and personality (Neuroticism–Extraversion–Openness Personality Inventory). Latent variable modeling was applied to obtain error-free indices of each factor, followed by voxel-based lesion–symptom mapping to elucidate their neural substrates. Regression analyses revealed that latent scores for psychometric intelligence reliably predict latent scores for cognitive flexibility. Lesion mapping results further indicated that these convergent processes depend on a shared network of frontal, temporal, and parietal regions, including WM association tracts, which bind these areas into an integrated system. A targeted analysis of the unique variance explained by cognitive flexibility further revealed selective damage within the right superior temporal gyrus, a region known to support insight and the recognition of novel semantic relations. The observed findings suggest an integrative framework for understanding the neural foundations of adaptive behavior, indicating that core elements of cognitive flexibility emerge from a distributed network of brain regions that support specific competencies for human intelligence.

Planning s0050

It has long been argued that patients with lesions in the PFC have difficulties in decision-making and problem solving in real-world, unstructured situations, particularly problem types involving planning. Recently, several researchers have questioned our ability to capture and characterize these deficits adequately using standard neuropsychologic test batteries and have called for tests that reflect real-world task requirements more accurately. One study (Goel et al., 1997) presented data from 10 patients with focal lesions to the PFC and 10 normal control subjects engaged in a real-world financial planning task. The results indicated that patients' performance was lower at a global but not local level. In other words, patients showed difficulty in organizing and structuring their plan's problem space. During problem solving, they had difficulty in allocating adequate effort to each problem solving phase. Patients also had difficulty when there were no right or wrong answers in real-world planning problems. They also had difficulty in generating their own feedback. For instance, they terminated the session before the details were shown and all the goals satisfied. Finally, patients did not take full advantage of the fact that constraints on real-world problems were negotiable. Taken together, these findings suggest that patients' deficits in real-world planning tasks involve the following areas: inadequate access to SECs, difficulty in generalizing from particulars, failure to shift between mental sets, and poor judgment regarding adequacy and completeness of a plan. p0160

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s0055 **Reasoning**

p0165 Analogic reasoning is at the core of the generalization and abstraction processes that enable concept formation and creativity. The few studies that have been performed on patients have highlighted the importance of the PFC in analogic reasoning. A recent study (Urbanski et al., 2016) examined analogic reasoning abilities in 27 patients with focal damage in the PFC and performed voxel-based lesion–behavior mapping and tractography analyses to investigate the structures critical for analogic reasoning. The findings revealed that damage to the left rostrolateral PFC region impaired the ability to reason by analogy. A short version of the analogy task predicted the existence of a left rostrolateral PFC lesion with good accuracy. Experimental manipulations of the analogy tasks suggested that this region plays a role in relational matching or integration. These results also suggested that analogy tasks should be translated to clinical practice to refine the neuropsychologic assessment of patients with PFC lesions.

s0060 **Problem solving**

p0170 The PFC has long been thought to play an important role in problem solving behavior. Clinical studies have described patients who show marked impairments in everyday life, including planning problem solving and decision-making. Several factors contribute to such impairments, including difficulty in generating possible problem solutions and then selecting an appropriate solution. Channon and Crawford (1999) described the performance of participants with unilateral anterior or posterior lesions compared to healthy controls in their ability to solve real-life problems. These covered a range of everyday interpersonal situations and were presented both in video and story format. Participants also carried out a set of more abstract neuropsychologic tests. Individuals with brain lesions showed impairment relative to controls in both everyday problem solving and on more abstract tests involving EF and memory. The anterior group was impaired on more aspects of everyday problem solving than the posterior group showing reduced ability in generating possible solutions and also impairment in selecting appropriate problem solutions.

p0175 Recently, Barbey et al. (2014) investigated the neural bases for social problem solving (measured by the Everyday Problem Solving Inventory) and examined the degree to which individual differences in performance are predicted by a broad spectrum of psychologic variables, including psychometric intelligence (measured by the Wechsler Adult Intelligence Scale), emotional intelligence (measured by the Mayer, Salovey, Caruso Emotional Intelligence Test), and personality traits (measured by the Neuroticism–Extraversion–Openness Personality Inventory). Scores for each variable were obtained, followed by voxel-based lesion–symptom mapping. Stepwise regression analyses revealed that working memory, processing speed, and emotional intelligence predicted individual differences in everyday problem solving. A targeted analysis of specific everyday problem solving domains (involving friends, home management, consumerism, work, information management, and family) revealed psychologic variables that selectively contribute to each. Lesion mapping results indicated that social problem solving, psychometric intelligence, and emotional intelligence are supported by a shared network of frontal, temporal, and parietal regions, including WM association tracts that bind these areas into a coordinated system. The results support an integrative framework for understanding social intelligence and make specific recommendations for the application of the Everyday Problem Solving Inventory to the study of social problem solving in health and disease.

EF neuroimaging evidence s0065

PFC s0070

In the past 2 decades, numerous studies used functional neuroimaging techniques (including PET and fMRI) to identify cerebral areas involved in specific EF tasks. In general, neuroimaging studies have provided extensive evidence for an association between executive functioning and the frontal lobes, and helped to identify subdivisions of the human PFC. The issues of whether different regions of the PFC are specialized for different aspects of EF is central to gaining a better understanding of executive processing through neuroimaging (Elliott, 2003). p0180

The vlPFC is proposed to control the retrieval of representations from the posterior cortex and, according to some (D'Esposito et al., 1998), the online maintenance of these accessed representations. The dlPFC region has been most commonly implicated in manipulating information from these representations. For example, although dlPFC is probably not involved in processes such as remembering a telephone number, it would play a role if one tried to dial the number in reverse order (requiring the deliberative manipulation of information). The dlPFC has been repeatedly shown to be involved in complex functions such as making plans for the future (Gilbert and Burgess, 2008). p0185

Working memory s0075

Numerous functional neuroimaging studies have provided a wealth of information about the likely organization of working memory processes within the human lateral frontal cortex (Owen, 1997). Unlike lesion p0190

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studies, neuroimaging evidence has documented specific associations between working memory tasks and activation of the dlPFC or vlPFC. For example, in one of the pioneering fMRI studies in this field (D'Esposito et al., 1995), participants were asked to perform semantic and visuospatial tasks, either alone (single task) or simultaneously (dual task). When single-task conditions were compared to dual-task conditions all subjects showed a significant increase in activation bilaterally in the dlPFC, even though neither task activated PFC when performed alone. This data provides critical support to the specific hypothesized role of the dlPFC in working memory.

p0195 Extensive attempts were made to associate different working memory processes with specific anatomic locations. This proved to be a challenging task, with different attempts to summarize existing data producing different conclusions (Baddeley, 2003). For example, Smith and Jonides (1998) reported different patterns of brain activation between processing verbal information (left-hemisphere posterior parietal cortex, temporal cortex and premotor cortex, and supplementary motor areas) and spatial information (right hemisphere posterior parietal, occipital, and frontal cortex). The authors argued that verbal storage and rehearsal processes can be anatomically dissociated. However, a more recent meta-analysis (Wager and Smith, 2003a) could not confirm this finding. Analyzing 60 neuroimaging studies of working memory, and focusing on interactions between different types of information to be remembered (spatial, verbal, and object) and different types of EF (continuous updating of WM, memory for temporal order, and manipulation of information in WM), the authors report no dissociation between spatial and nonspatial storage in the frontal cortex.

p0200 A more recent meta-analysis (Owen et al., 2005) found that the during the *n*-back task (described earlier) there is increased activation in the lateral premotor cortex, dorsal cingulate and medial premotor cortex, dlPFC and vlPFC, frontal poles and medial and lateral posterior parietal cortex.

s0080 Inhibitory control

p0205 Normal human behavior and cognition are tightly associated with the ability to inhibit inappropriate thoughts, impulses, and actions. Typically, inhibition in a laboratory setting is operationalized as the suppression of a prepotent response, and different imaging studies report an association between response inhibition and brain activation of primarily frontal and largely right hemisphere regions (Garavan et al., 1999).

p0210 For example, various studies used the stop signal task, in which participants are instructed to press a button as fast as possible when presented with one (frequent) target and to avoid any response when presented with another (rare) target. Imaging research with this task suggests that the right inferior frontal cortex is a critical region for stop signal response inhibition. Moreover, it was argued that, in the stop trials, responses that are already initiated can be rapidly reversed through involvement of both frontal and basal ganglia-related circuits (Aron et al., 2007).

Another study used event-related fMRI to identify p0215
cortical regions that are activated during withholding of a prepotent motor response. Regions identified in this study were strongly lateralized to the right hemisphere and included the middle and inferior frontal gyri, frontal limbic area, anterior insula, and inferior parietal lobe (Garavan et al., 1999).

Another study used event-related fMRI to study the p0220
development of inhibitory control. The authors report that children had more difficulty inhibiting a response to stop trials, and successful response inhibition was associated with stronger activation of prefrontal and parietal regions. These findings suggest that behavioral inhibition improves across childhood and is associated with the maturation of underlying ventral frontostriatal circuitry (Durston et al., 2002).

More recently, Munakata et al. (2011) argued for two p0225
separate forms of inhibitory mechanisms: *directed global form*, or prefrontal regions providing contextual information relevant to when to inhibit all processing in a subcortical region; and *indirect competitive form*, or prefrontal regions providing excitation of goal-relevant options. These distinctions may be crucial for understanding the mechanisms of inhibition and how they can be impaired or improved.

Cognitive flexibility s0085

Recent reviews of cognitive flexibility have identified a p0230
distributed network of frontoparietal regions involved in flexible switching, including high-level cortical association areas (vlPFC, dlPFC, anterior cingulate, and right anterior insula), the premotor cortex, the inferior and superior parietal cortices, the inferior temporal cortex, the occipital cortex, and subcortical structures such as the caudate and thalamus (Dajani and Uddin, 2015). Ongoing work is attempting to understand how these brain regions interact to form a coherent network to implement cognitive flexibility.

For example, in one study (Badre and Wagner, 2006) p0235
event-related fMRI was performed during a task-switching paradigm (i.e., the participants were required to switch between two simple cognitive tasks). Brain activation was recorded at left vlPFC areas as well as inferior parietal cortex, suggesting that the left vlPFC cortex resolves the conflict during task switching to facilitate flexible performance. Another study (Lie et al., 2006) used fMRI to segregate different network

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components in the frontoparietal and striatal regions underlying the Wisconsin Card Sorting Test. Additionally, activations in the cerebellum were found to be related to set-shifting.

s0090 Planning and problem solving

p0240 Recent functional neuroimaging studies have identified the dlPFC as the critical area for performance on the Tower of London planning task. For example, an event-related fMRI study used a variant of the Tower of London task to identify underlying brain circuitry (Fincham et al., 2002). The results of this study suggest a functional dissociation between a right prefrontal–parietal network whose activity reflects the working memory demands of the task and a left posterior dlPFC component associated with the active selection of task appropriate subgoals in the service of planning.

p0245 Another study looked at the hemispheric differences in PFC as they relate to planning (Newman et al., 2003). The results from this study suggest that both the left and right prefrontal cortices are equally involved during the solution of the problem, but the right PFC is differentially involved in constructing the plan for solving the problem, whereas the left PFC is involved in supervising the execution of the plan.

p0250 An fMRI study reported a double dissociation between medial and lateral anterior PFC in planning (Koechlin et al., 2000). The authors showed that medial anterior PFC, in association with the ventral striatum, was engaged preferentially when participants executed tasks in sequences that were expected. However, the polar PFC, in association with the dorsolateral striatum, was preferentially involved when participants performed tasks in sequences that were unpredictable.

p0255 To conclude, a number of cortical and subcortical regions not located in PFC were also activated by the versions of the Tower of London task, including the caudate nucleus, the presupplementary motor area, the anterior premotor cortex, the posterior parietal cortex, and the cerebellum (Unterrainer and Owen, 2006). The available functional neuroimaging data suggest that, while the dlPFC plays a critical role in planning, it does so through functional interactions with multiple cortical and subcortical regions.

s0095 Reasoning

p0260 Over the course of the past decade, contradictory claims have been made regarding the neural bases of reasoning. A quantitative meta-analysis of 28 neuroimaging studies of deductive reasoning published between 1997 and 2010, combining 382 participants, identified consistent areas of activations across studies in neural systems located in both cortical (frontal and parietal cortices) and subcortical (left basal ganglia) structures (Prado et al., 2011).

For example, one study (Christoff et al., 2001) used event-related fMRI to study brain activation during problems adapted from the Raven's Progressive Matrices. This study found activation in the PFC that was specific to when two dimensions of variation need to be considered simultaneously and integrated, namely relational integration. Furthermore, the process of relational integration was specifically associated with bilateral rostrolateral PFC and right dorsolateral PFC. p0265

In summary, both lesion and imaging studies provide evidence for a strong association between the PFC and EF, with specific aspects of EF linked to specific PFC areas (see Fig. 11.1 and Table 11.3). As a caveat, severe EF impairments are sometimes observed with no frontal lobe brain damage, but instead may be caused by damage to one of the many cortical and subcortical regions that interact with the PFC (e.g., frontostriatal circuits). Moreover, frontal brain injury can sometimes result in minimal or no executive deficits, reflecting the role of other factors such as environment (e.g., socioeconomic status; see section "Development of EF") or genetics (see section "EF and genetics") on EF performers. p0270

NEUROPSYCHOLOGIC TOOLS TO ASSESS EF s0100

Many neuropsychologic tests were developed over the years to assess EF and provide age- and education-based norms, which allow standardized interpretation of the achieved score. Apart from standardized neuropsychologic tests, other measures can and should be used, such as behavior checklists, observations, interviews, and work samples. An ideal assessment of EF involves gathering data from several tests and sources and combining the information to look for shared patterns across time and settings. p0275

Standard tests s0105

Several neuropsychologic tests assess EF, the most used and studied being the Wisconsin Card Sorting Test, i.e., assessing the ability to learn and shift rules (Milner, 1963); the Trail Making part B, i.e., assessing the ability of set-shifting (Armitage, 1946); the Stroop Color Word Interference Test, i.e., assessing the ability of inhibition of irrelevant information (Perret, 1974); and the Tower of Hanoi, i.e., assessing planning abilities. In addition, several test batteries allow a wider assessment of several EF aspects, e.g., Delis–Kaplan Executive Function System (D-KEF) and the Behavioral Assessment of Dysexecutive Syndrome (BADS). What these tasks generally have in common is that they require processing beyond the well-learned associations between stimuli and responses. Table 11.4 provides a summary of major neuropsychologic assessment tools. p0280

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t0020 *Table 11.3*

Summary of EF functional neuroanatomy based on lesions and neuroimaging EF evidence.

	Lesion studies		Neuroimaging studies	
Function	Main task tested	Associated brain region	Main task tested	Associated brain region
Working memory	Verbal and spatial memory span	Lateral PFC (Owen et al., 2005)	Manipulation information: reverse span, dual tasks	dlPFC (Cristofori et al., 2015)
	N-BACK	Medial OFC (Barbey et al., 2011)		
Inhibitory control	Comprehensive battery of neuropsychologic tasks	dlPFC and ACC (Gläscher et al., 2009)	Stop signal task	Right inferior frontal cortex (Aron et al., 2007) and ventral frontostriatal circuitry (Durston et al., 2002)
Cognitive flexibility	Discrimination learning task	OFC (Schoenbaum et al., 2009)	Wisconsin Card Sorting Test (WCST)	Frontoparietal and striatal regions (Lie et al., 2006)
	Category switching, letter fluency, and category fluency	Right superior temporal gyrus (Barbey et al., 2013a, b)		
Planning	Real-world financial planning task	PFC (Goel et al., 1997)	Tower of London	Left posterior dlPFC interacting with multiple cortical and subcortical regions (Fincham et al., 2002)
Reasoning	Analogy task	Left rostrolateral PFC (Urbanski et al., 2016)	Raven's Progressive Matrices (Relational integration)	Bilateral rostrolateral PFC and right dorsolateral PFC (Christoff et al., 2001)

t0025 *Table 11.4*

EF assessment tools.

Test	EF component
Clinical rating/bedside assessment	
Cambridge Neurological Inventory (Chen et al., 1995)	Motor initiation, sequencing, and inhibition
Lab-based/constraint environment	
WCST (Heaton et al., 1993)	Switching, perseveration
Verbal Fluency Test (Lezak, 1995)	Verbal production
Design Fluency Test (Jones-Gotman and Milner, 1977)	Nonverbal switching
Stroop Test (Stroop, 1935)	Verbal inhibition
Hayling Sentence Completion Test (Burgess and Shallice, 1996)	Verbal inhibition
Brixton Spatial Anticipation Test (Burgess and Shallice, 1996)	Rule detection and impulsivity
Tower of London (Shallice, 1982) and Tower of Hanoi (Humes et al., 1997)	Planning
Sustained Attention to Response Task (Robertson et al., 1997)	Sustained attention and inhibition
Six Elements Test (Wilson et al., 1996)	Planning, strategy allocation
Greenwich Test (Burgess et al., 2000)	Executive memory, planning, and intentionality
Naturalistic environment	
Multiple Errands Test (Shallice and Burgess, 1991)	Strategy allocation, planning
Iowa Gambling Task (Bechara et al., 1994)	Emotion and decision-making
Assessment of Motor and Process Skills (Fisher, 1993)	Motor and process skills
Frontal Systems Behavior Scale (Chiaravalloti and DeLuca, 2003)	Initiation and disinhibition

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s0110
Ecologic tests

p0285 The challenge in evaluating EF derives not only from determining the appropriate laboratory tests to administer, but also from understanding EF demands in more ecologic settings. Several measures address the need for ecologic evaluations of EF, but most of them require the input of a caregiver to compare to self-report by the TBI patient. For example, the Frontal Systems Behavior Scale (FrSBe) (Grace and Malloy, 2001) evaluates apathy, disinhibition, and executive dysfunction, and is considered a good predictor of TBI patient community reintegration (Reid-Arndt et al., 2007). Although a lower score on this test predicted less community reintegration, other standard neuropsychologic tests were unrelated to community reintegration.

p0290 Another ecologically valid questionnaire is the Brock Adaptive Functioning Questionnaire (BAFQ) (Dywan and Segalowitz, 1996). The BAFQ evaluates the frequency of EF deficits in 12 domains: planning, initiation, flexibility, excess caution, attention, memory, arousal level, emotionality, impulsivity, aggressiveness, social monitoring, and empathy. The BAFQ was shown to distinguish between individuals with TBI who might return to their preinjury employment level from those who might require employment reassignment (Simpson and Schmitter-Edgecombe, 2002).

p0295 Ecologic measures of EF can complement an objective test-based evaluation to better estimate the patient's potential to resume everyday adaptive functioning. Moreover, along with standard neuropsychologic assessments, an ecologic evaluation can help to define more tailored interventions to improve EF.

s0115
Ecologic validity

p0300 Once an executive impairment is documented using neuropsychologic tests in the clinic, it is still not clear how it will affect one's daily life. Even the most ecologically valid tests described here are quite unlike most real-life situations. For example, it's not clear what kind of difficulties a patient with poor performance on the WCST will experience in daily life (Burgess et al., 2006).

p0305 A comprehensive study of the ecologic validity of EF measures (Burgess et al., 1998) examined associations between performance on neuropsychologic tests as well as self and caregiver reports of dysexecutive symptoms (DEX) in a sample of neurologic patients ($n = 92$) and controls ($n = 216$). Results indicated significant correlations between informant-reported DEX symptoms and neuropsychologic test performance. However, associations were not specific to tests of EF and a few other cognitive tests evaluating intelligence, visual perception and memory were also correlated with DEX ratings. EF tests were more specifically associated with patients' lack of insight, as reflected in differences between their self-report DEX ratings and informant reports.

p0310 A later study (Ready et al., 2001) found that neuropsychologic measures of EF were significant predictors of achievement and work-related behavior, whereas subjective rating scales were more strongly associated with disinhibited, risk-taking, and aggressive behaviors.

p0315 More recently, Guevara et al. (2016) focused on the level of burden reported by caregivers of individuals with dysexecutive symptoms 40 years after TBI. Findings from this study reveal that impaired EF due to lesions in the left dorsal anterior cingulate cortex (dACC) and the dlPFC were associated with higher levels of long-term caregiver burden. Participants with lesions in these areas demonstrated deficits in cognitive and behavioral indicators associated with dysexecutive syndrome but not in language abilities, space and object perception, memory, depression or posttraumatic stress. Considering these findings, clinicians should be aware that a caregiver whose spouse/offspring/parent has a left dACC/dlPFC lesion is at higher risk of feeling burdened and should refer caregivers to available resources.

TRAINING TO IMPROVE EF s0120

p0320 Developing intervention plans to improve EF is a highly desired goal, and many different tools and protocols were examined for their effectiveness. Overall, despite intensive cognitive training, a large majority of patients with dysexecutive syndrome fail to fully regain normal EF. This section describes the current tools and future possibilities of EF training.

Cognitive rehabilitation of EF s0125

p0325 Cognitive rehabilitation of EF is crucial, since this cognitive domain significantly affects everyday life and social functioning. To date, cognitive rehabilitation has aimed at improving specific EF abilities, such as planning, inhibition, or updating, rather than the entire EF domain.

p0330 Some EF training studies focus on awareness of EF deficits during treatment. For example, Toglia et al. (2010) conducted a single-subject trial design, to promote strategy use across situations and increase self-regulation, awareness, and functional performance. Sessions were divided into three phases: error discovery, strategy, and reinforcement. All participants demonstrated positive changes in self-regulation and strategy use. These results provide support for the feasibility of cognitive rehabilitation therapies to enhance functional performance and awareness. However, larger controlled studies are needed to confirm or qualify these observations.

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p0335 Another group of EF training studies did not take deficit awareness into account. For instance, the efficacy of cognitive-motor dual-task training was assessed as a mean to improve EF in TBI patients. This treatment included exercises such as walking, in combination with tasks of increasing cognitive load. The "treatment" group was compared to a "no treatment" group. The primary outcome measure was a task requiring participants to walk and carry out a spoken sentence. Secondary outcome measures involved performing either two motor tasks or two cognitive tasks. The findings revealed better performance on the primary outcome measure, but little generalization to other measures in the treatment group (Evans et al., 2009). Similarly, Hewitt et al. (2006) assessed the ability of TBI patients to develop a plan to accomplish a familiar task such as planning a trip. The outcome measures were number of steps listed and effectiveness of the plan. These measures were assessed after the training by blinded raters, and both groups improved. However, the strategy-training group improved more from pre- to posttraining. This study indicates that combining learning and using strategies is useful in improving complex planning.

p0340 Individuals with EF impairments have relatively preserved verbal knowledge. However, they may fail to use this knowledge to guide goal-oriented behaviors. Goal Management Training (GMT) is a well-established rehabilitation technique focused on goal-directed behaviors. The aim of this training is to monitor and adjust goals during ongoing behavior (Levine et al., 2000). This approach is based on a theory of sustained attention (Robertson and Garavan, 2000), which claims that the right fronto-thalamic-parietal network provides the neural support for ongoing activation of attention, which allows for the maintenance of higher-order goals in working memory. Impairments in sustained attention may determine a displacement of higher-order goals (e.g., preparing a dinner or posting a letter), resulting in inadequate and distracted behaviors. The GMT consists of five steps: (1) orient awareness toward the actual state of the situation; (2) define the goal of the task; (3) list subgoals; (4) learn the subgoals; and (5) check if the result of an action corresponds to the stated goal. In the case of a discrepancy, the steps are repeated. Levine and collaborators (2000) assessed the effectiveness of GMT in brain-injured patients. Participants were randomly assigned to GMT or motor skills training. Only participants enrolled in the GMT improved on paper-and-pencil tasks. This study provided support for the efficacy of GMT for improving EF, and these results were later replicated (Levine et al., 2011). The authors compared an extended version of GMT to an alternative intervention, the Brain Health Workshop. The findings provided evidence of the beneficial effects of GMT on sustained attention and visuospatial problem solving, reflecting a generalization effect of the training (Levine et al., 2011).

From EF rehabilitation to social reintegration s0130

p0345 EF deficits reduce individuals' abilities to regain satisfactory social lives. Understanding the processes that promote social reintegration and involve EF deficits is challenging. Training in goal management, the core of EF rehabilitation, may treat deficits in EF and help individuals regain social activities. In an attempt to test this argument, Libin et al. (2015) developed the COmmunity ParticipAtion through Self-Efficacy Skills Development program (COMPASS goal). This is a novel patient-centered intervention for community reintegration of veterans with mild TBI. COMPASS goals are based on the principles of goal self-management. Goal setting is a core skill in self-management training by which individuals learn to improve their abilities. Over a 3-year period, this ongoing project plans to recruit 110 participants with EF impairments at least 3 years postinjury. This program has strict inclusion criteria combining both clinical diagnosis and standardized scores that are >1 SD from the normative score on the Frontal Systems Behavior Scale (FrSBE) (Grace and Malloy, 2001). Participants will be randomly allocated to one of two groups: goal management (training) and supported discharge (control). The training is administered in eight sessions and assessments occur before, right after, and 3 months after training. To date, the COMPASS goal has begun testing and has enrolled 20% of its target. A structured approach to goal self-management such as the COMPASS program may foster greater independence and self-efficacy as well as help individuals reach realistic goals.

Noninvasive brain stimulation: TMS and tDCS s0135

p0350 Noninvasive brain stimulation may enhance EF recovery/training through the induction and enhancing of plasticity processes. Emerging research suggests that noninvasive brain stimulation tools, such as transcranial magnetic stimulation (TMS) and transcranial direct current stimulation (tDCS) in association with cognitive training may enhance recovery of EF.

p0355 In TMS, a small coil held near the scalp generates a magnetic field that produces an electric field in the underlying brain areas. With the appropriate frequency and intensity of stimulation, TMS can be used to excite the specific brain regions. Reshaping abnormal postinjury neuronal activity might be an effective strategy to facilitate clinical rehabilitation effects. In a study with rodents, high-frequency TMS was delivered twice a week over a 4-week period to rats to see if it could restore

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neuronal activity and improve neurophysiological and behavioral functions (Lu et al., 2015). The findings showed that TBI rats that underwent TMS therapy had increased evoked cortical responses (189%), evoked synaptic activity (46%), evoked neuronal firing (200%), and increased expression of cellular markers of neuroplasticity in their noninjured brain areas compared to brain-injured rats that did not receive therapy. Notably, these rats showed less hyperactivity on behavioral tests. Earlier studies have shown similar effects in humans: for example, high-frequency repetitive transcranial magnetic stimulation (rTMS) was used to excite dlPFC neurons in healthy participants during a task-switching paradigm (Vanderhasselt et al., 2007). This paradigm required switching between two conditional responses with mutually incompatible response-selection rules. The results showed that reaction time on cued switching trials decreased with rTMS, as compared to noncued switch trials. No changes occurred after the sham condition where no magnetic stimulation was provided. These promising findings suggest that in the future cognitive rehabilitation protocols might include rTMS to improve EF performance.

p0360 Another noninvasive brain stimulation method, tDCS, uses low-amplitude direct current to modify neuronal activity. Anodal or cathodal direct current polarization to induce neural firing changes has been used since the 1960s. Anodal tDCS elicits prolonged increases in neural excitability and facilitates regional brain activity, while cathodal tDCS elicits the opposite effects (Nitsche and Paulus, 2001). While the induced effects last for the period of the stimulation, repeated stimulations reportedly result in effects lasting several weeks, a crucial feature with regard to cortical plasticity (Fregni et al., 2006; Boggio et al., 2007).

p0365 Recently, Dockery et al. (2009) studied the effects of tDCS of the left dlPFC on planning. They assessed performance on the Tower of London task during and after anodal, cathodal (1 mA, 15 min), and sham tDCS in 24 healthy participants. The results revealed a double dissociation of polarity and training phase. Better planning performances were found with cathodal tDCS during acquisition and early consolidation, and with anodal tDCS in the later sessions. The findings indicated that both anodal and cathodal tDCS could improve planning performance. These data demonstrated training-phase-specific effects of tDCS. The authors concluded that the excitability supported by decreasing cathodal tDCS mediates the early beneficial effect through noise reduction of neuronal activity, while the excitability supported by anodal tDCS is mediated by adaptive configurations of specific neuronal connections. The improvement in planning was maintained at 6 and 12 months after training. The specific coupling of stimulation and training-phase interventions encourages its application to treating individuals with brain injury.

The fact that anodal tDCS stimulation often produces p0370
an opposite outcome compared to cathodal tDCS has been recently exploited by Westwood and Romani (2018). The authors showed that applying anodal tDSC to inferior frontal gyrus did not increase healthy participants' performance in working memory and verbal fluency.

The use of noninvasive brain stimulation to modulate p0375
EF requires further technical study to determine the duration of any observed effects and the optimal dosage needed to produce long-lasting beneficial effects. Taken together, TMS and tDCS studies indicate that noninvasive brain stimulation might serve in the future as a complementary tool for improving EF in healthy individuals and patients.

DEVELOPMENT OF EF s0140

The PFC is known to mature slowly, with some parts p0380
continuing to develop through adolescence and into adulthood (Gilbert and Burgess, 2008). Similarly, EFs gradually develop and are among the last mental functions to reach maturity. This development of the EF is believed to account for many of the differences perceived between the stages of cognitive development across the life span. On average, children and older adults both show poorer EF performance relative to young adults. This section reviews the major developmental stages and their association with EF.

During childhood and adolescence, EFs are continu- p0385
ously improving (Jurado and Rosselli, 2007). Infants are mainly stimulus-bound and react to the present and to events in their immediate surroundings. However, even during the first year of life infants can show evidence of extracting basic rules, as well as attention and memory abilities (Rose et al., 2016). These basic information processing abilities are believed to provide the foundation for the later development of executive functioning. By the child's first year the ability to inhibit overlearned behavior is developed, allowing the child increased attentional control over the environment. Skills such as planning and set-shifting emerge by age 3, with significant improvement after age 7. The ability to inhibit task-irrelevant information shows its greatest development later between the ages 6 and 10.

There are individual differences in the rate and level of p0390
EF improvement across childhood, which can be accounted for by genetic factors (see section "Development of EF") and by environmental factors. In one study parental socioeconomic status predicted the children's working memory performance at age 4.5 years and planning by first grade (Hackman et al.,

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2015). These differences in performance largely persisted through middle to late childhood, suggesting that early development of EF is affected by early childhood socioeconomic status.

p0395 While many studies have investigated the development of EF skills in early and late childhood, only a few studies have investigated these changes during adolescence. There is evidence that performance on tasks including inhibitory control, processing speed, working memory, and decision-making continue to develop during adolescence. For example, improvement during adolescence was observed on tasks of selective attention, working memory, and problem solving. There is also evidence of nonlinear development throughout adolescence with poorer performance on EF tasks in late adolescence compared to early/middle adolescence, possibly due to neural reorganization (Blakemore and Choudhury, 2006). For example, a recent longitudinal study that followed participants aged 17, 18, and 19 for 12–16 months reported decline over this age range on a sorting test (Taylor et al., 2015). In general, across adolescence most EFs improve or stabilize, while a few may show decline.

p0400 In the same way that maturation of the PFC has been connected to the development of EFs, the decline in EFs at the other end of the life span has been associated with anatomic changes in the brain during normal aging. The human brain is believed to undergo a gradual reduction in volume that begins during middle adulthood. Some brain areas known to be vulnerable to the effects of aging are the hippocampus and prefrontal regions (Raz et al., 2005). While increasing age has been found to correlate with decline in the performance of EF, it has often been difficult to differentiate this decline from a general decline in cognitive abilities (Crawford et al., 2000). A well-documented example is the age-related changes in executive attentional control and inhibition, which manifest with increased vulnerability to interference and working memory decline (Hasher and Zacks, 1998). One study, focused on the aging of planning ability, reported that younger adults (mean age of 22.7 years) outperform their older adults counterpart (68.1 and 78.75 years) when tested on the Tower of Hanoi task (Sorel and Pennequin, 2008). In a meta-analysis of performance on the WCST, older adults were found to complete fewer categories and commit more perseverative errors (Rhodes, 2004). These results indicated that robust age differences are present on standard EF tests.

p0405 However, while some cognitive aging theories stress the effect of executive decline throughout adulthood, others argue for a modest role for executive control as an explanatory mechanism for cognitive aging. Importantly, executive control (as defined by resistance to interference and task shifting) does not explain any age-related variance in complex cognition over and beyond the effects explained by a slower speed of processing (Verhaeghen, 2011).

EF AND GENETICS s0145

EF may be influenced by genetic polymorphisms. The p0410
following section reviews the major studies that made a significant contribution to the relationship between EF and genetics, with particular focus on key genes that seem to play a dominant role in EF regulation: 5-HT, COMT, BDNF, and APOE.

5-HT s0150

Twin studies have established that genetic influences p0415
may account for a significant amount of variation in EF—especially for inhibition, shifting, and monitoring/working memory—ranging from 43% to 77% (Coolidge et al., 2000; Ando et al., 2001; Kuntsi et al., 2006). One candidate gene associated with EF is serotonin (5-HT) (see Logue and Gould, 2014). The role of 5-HT in the development of EF is related to the expression of 5-HT in the PFC (Puig and Gulledge, 2011). Serotonergic receptors are expressed in the PFC, which control 5-HT activity (Enge et al., 2011). Variations in extracellular 5-HT in the PFC have been associated with performance in response inhibition, reversal learning tasks, and other EFs in humans (Cools et al., 2008) and nonhuman primates (Walker et al., 2006). Given the prevalence of 5-HT regulation and EF performance in general, the functional polymorphism in the promoter region of the 5-HT transporter gene (5-HTTLPR) is a plausible candidate for EF functioning (Hu et al., 2006). A recent study from Li et al. (2015) found a significant interaction between 5-HTTLPR and parental supervision for cognitive flexibility. More specifically, given low levels of parental supervision, adolescents with the L/L genotype had lower flexibility compared to adolescents with S/S or S/L genotypes. These findings showed that adolescents with poor parental supervision and with the L/L genotype were vulnerable to lower cognitive flexibility. Better understanding of this kind of vulnerability may help in targeting preventive intervention in adolescents.

COMT s0155

Dopamine plays a key role in the modulation of activities p0420
in the PFC during EF tasks (Williams and Goldman-Rakic, 1995; Seamans et al., 1998). These modulations have moved researchers to investigate genes influencing dopamine levels and their role in EF. Catechol-*O*-methyltransferase (COMT) is an enzyme that metabolizes catecholamine neurotransmitters, such as dopamine, epinephrine, and norepinephrine. The COMT gene is a

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regulator of dopamine levels in the PFC. Studies of COMT knockout mice revealed that memory performance had improved (Gogos et al., 1998). Similarly, in humans, higher concentrations of catecholamine in the brain can result in better working memory performance following brain injury. Specifically, the Val158Met polymorphism—a common COMT gene variation—is linked to cognitive control during working memory (Goldberg and Weinberger, 2004). While the Met allele is associated with a reduction in COMT enzyme activity, and subsequent increased levels of prefrontal dopamine, the Val allele is associated with high COMT enzyme activity, and lower levels of prefrontal dopamine. As a result, individuals with the Met/Met genotype showed better performance on working memory tasks compared to individuals with at least one Val allele. Individuals with the Met/Met genotype can respond better to working memory training compared to individuals with other polymorphisms. The Met allele could protect against the deterioration of EF following brain injury. Egan et al. (2001) found that the functional Val108/158 Met polymorphism in the COMT gene was associated with better EF performance on the WCST. In a more recent study, brain-injured patients with Met alleles also performed better on the WCST compared to Val allele patients (Lipsky et al., 2005). This finding suggested that the COMT gene could influence EF recovery after brain injury modulating the availability of PFC dopamine.

s0160
BDNF

p0425 Another gene that might influence EF recovery after brain injury is the brain-derived neurotrophic factor (BDNF). Following brain injury, the brain activates repair mechanisms and promotes neuroregeneration, which is facilitated by neurotrophic factors, including BDNF (Ray et al., 2002).

p0430 BDNF promotes survival and synaptic plasticity in the brain (Binder and Scharfman, 2004). Within nucleotide 196 of the BDNF gene there can be a G-to-A substitution, which results in a Val to Met (Val66Met) substitution at codon 66 (Chen et al., 2008). The Val66-Met BDNF polymorphism has been associated with cognitive functioning and clinical pathology (Bath and Lee, 2006). In healthy populations, the Met allele has been linked to impaired working memory (Hariri et al., 2003). However, Met allele carriers have a functional advantage over Val allele carriers in response inhibition tasks (Beste et al., 2010). Krueger et al. (2011) investigated the way in which the Val66Met BDNF polymorphism affects EF recovery after brain injury. Compared to controls with the Val allele, brain-injured participants with the Val allele performed worse on EF tests measured with the Delis–Kaplan Executive Function System battery. No differences in EF performance were found between patients with damage to the PFC and controls that had the Met allele. This study indicates that having the BDNF met allele is a good predictor of better EF recovery after PFC brain damage.

APOEϵ4 s0165

Another gene that has a key role in cognitive function is the apolipoprotein E allele 4 (APOEϵ4). APOE transports, recycles, and clears lipids in the brain (Mahley and Rall, 2000). Lipids have a role in maintaining neurologic integrity and in supporting neural repair after brain injury (Adibhatla and Hatcher, 2007). In addition, it has been observed that APOE has antioxidant properties, associated with the integrity of the blood–brain barrier following brain injury (Horsburgh et al., 2000). Thus, APOE has an essential role in early neurologic responses to brain injury. Recently, Padgett et al. (2016) examined the effect of the APOE gene on EF working memory and speed of processing tasks during the early period of recovery after brain injury. In addition, researchers sought to understand whether the APOE ϵ4—associated with neurologic disorders such as Alzheimer's disease—had a potential protective role following brain injury. The authors produced a model that predicted performance on all tasks (EF, working memory, and speed of processing), accounting for around 20% of the variance. p0435

Overall, genetic studies have started to identify individual differences in polymorphisms that could affect normal EF functioning as well as recovery following brain injury. Examining the influence of polymorphisms on cognitive outcome has the potential to aid in the triaging and treatment of brain-injured patients, and could lead to interventions based on individualized profiles. p0440

SUMMARY AND OUTSTANDING QUESTIONS s0170

EF encompasses a broad range of cognitive functions, which include planning, problem solving, flexibility, and cognitive controls, among others. These functions are mainly regulated by the PFC, but also other posterior and subcortical regions play an important role, especially in the integration of information and emotion. Further research is still needed to better understand how the subcortical regions are involved in the regulation of EF. p0445

Impairments in EF reduce a person's ability to return to work or school and to resume satisfactory social activities. Understanding the consequences of EF deficits is crucial for accurate diagnoses and tailored rehabilitation programs to help individuals achieve independent lives. It is evident that there are still gaps in our understanding of EF impairments and prognosis. Longitudinal studies p0450

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t0030 *Table 11.5*

Outstanding questions.

Outstanding questions
What is the interplay between EF and social cognition?
What is the role of subcortical regions involved in EF regulation?
How long do EF training effects last?
Can improvement of EF have beneficial effects on society?
Which daily activity can improve EF (arts, music, care of an animal or person)?
Will we be porting EF to autonomous agents (robots or devices), reducing our dependence on the frontal lobes for these functions?

investigating the effect of EF impairments following brain injuries are crucial since they provide information about whether EF recovery facilitates functional and social outcomes. Ecologically valid studies can also be invaluable in determining the extent to which deficits measured in laboratory settings can be generalized to real-world functioning. To this aim it would be fundamental to understand how long EF training effects last and how long training protocols need to last in order to produce meaningful effects. Moreover, genetic profiles can provide important information concerning premorbid influences on the rehabilitation of EF as well as epigenetic capacities associated with rehabilitation interventions.

p0455 Since EFs are highly connected to our social functioning, investigations should be made into whether EF training could have positive effects on social interactions and, conversely, which daily activity to perform in order to improve EFs, such as arts, music, or care of an animal or a person.

p0460 Another important and emergent social change is the increasing use of autonomous agents in our daily life, such as robots and devices. How will the use of these agents modify our EF abilities and our use of our frontal lobes? See Table 11.5 for a summary of outstanding questions.

p0465 EF is a complex cognitive function, directly affecting key aspects of our daily behavior. From the studies reviewed here, discrepancies were found between lesion and neuroimaging studies, mostly due to different tasks/neuropsychologic assessments used to investigate EF. Our knowledge of the range and kind of EF processes would obviously benefit if convergent evidence on EF depended upon more similar tasks. Nevertheless, both lesion and imaging studies provide evidence for a strong association between the PFC and EF, with specific aspects of EF linked to specific PFC areas. Nevertheless, EF deficits are sometimes observed with lesions outside the PFC if the lesions affect one of the many cortical and subcortical regions that interact with the PFC. In addition, frontal brain injury can sometimes result in minimal or no EF deficits, reflecting the role of other factors such as environment (e.g., socioeconomic status) or genetics on EFs.

A final and rather puzzling observation about the nature and brain bases of EF remains troubling. Our review indicated that several PFC regions were associated with EF processes based on lesion mapping and neuroimaging studies. These EF processes include working memory, reasoning, valuation, inhibition, and a few more similar processes. Cognitive and neuroscience research has, for over a century, suggested that occipital, parietal, and temporal cortices represent aspects of information in networks based on the characteristics of the stored stimuli, whether they be the geometric properties of visual stimuli, the frequency of words, or functionality of an object (e.g., faces vs kitchen utensils). As one can imagine, there are many exemplars of each kind of stimulus category stored in posterior cortices enriching the neural networks that evolved for their representation. Yet, with the exploration of the PFC, most of the investigators of the PFC view it as a location for the range of EFs. There are simply not that many EFs, which suggests that the human brain has evolved a considerable number of neural and nonneural resources to support what appears to be a few valuable processes. It also has left a schism between the generally accepted view that the long-term storage of words, objects, and even social concepts occurs in nonfrontal regions and the apparent lack of cortical real estate that is committed to forms of knowledge such as schemas, themes, plans, and similar categories of more abstract knowledge. Do such long-term representations exist? If so, and given the kinds of impairments that are observed after damage to the PFC, are there networks within the PFC committed to their storage? And if that is the case, are they related to the EFs described in this chapter? Resolving this dilemma should improve our understanding of the genetic, morphologic, cognitive/social, and environmental factors that led to the evolution of advanced human behaviors and may provide landmarks from which we can steer humankind into the future. p0470

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